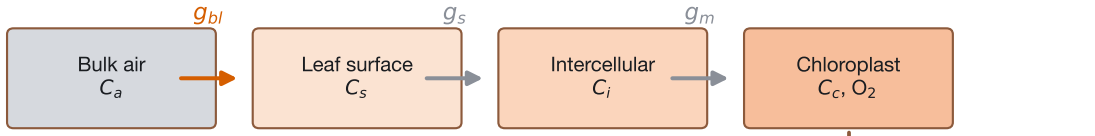
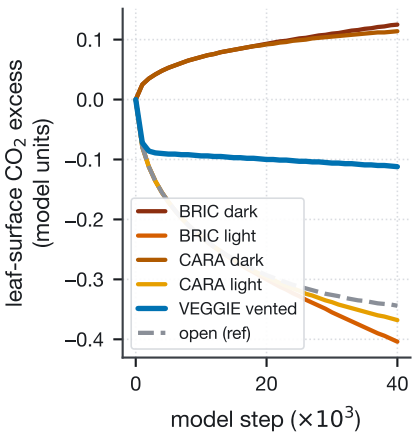
Growth hardware, not gravity, sets the atmosphere at the leaf

a The conductance chain

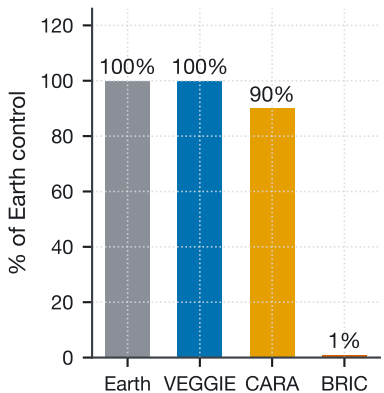


A sealed, lit enclosure lowers C_a itself — upstream of every conductance in the chain.

b Leaf-surface CO_2, microgravity



c Carbon gained over 12 h



d How fast a sealed canister changes

BRIC lit: CO_2 400 → ~0 ppm
7 minutes

BRIC dark: CO_2 → 1%
9.4 hours

BRIC dark: O_2 21% → 5%
6.5 days

CARA / VEGGIE: vents to cabin
bounded

The 7-minute drawdown is a mass balance, so it happens in the ground control too — which is why it cancels out of flight-vs-ground.

Gas transport from LunarLeaf-CFD (validated D2Q9 LBM solver); all hardware runs are in microgravity.